

Kepler exoplanets

the 2026 refresh

Kepler Objects of Interest: honest models, the habitable zone done on insolation, and an app that runs in your browser.

Rich Anderson • August 2026

Refresh of the 2020 Columbia Engineering data-analytics bootcamp team project
2020 team: Rich Anderson (ML pipeline), Damien Corr, Priscilla Lin, Tom Greff

Code, notebooks, research log

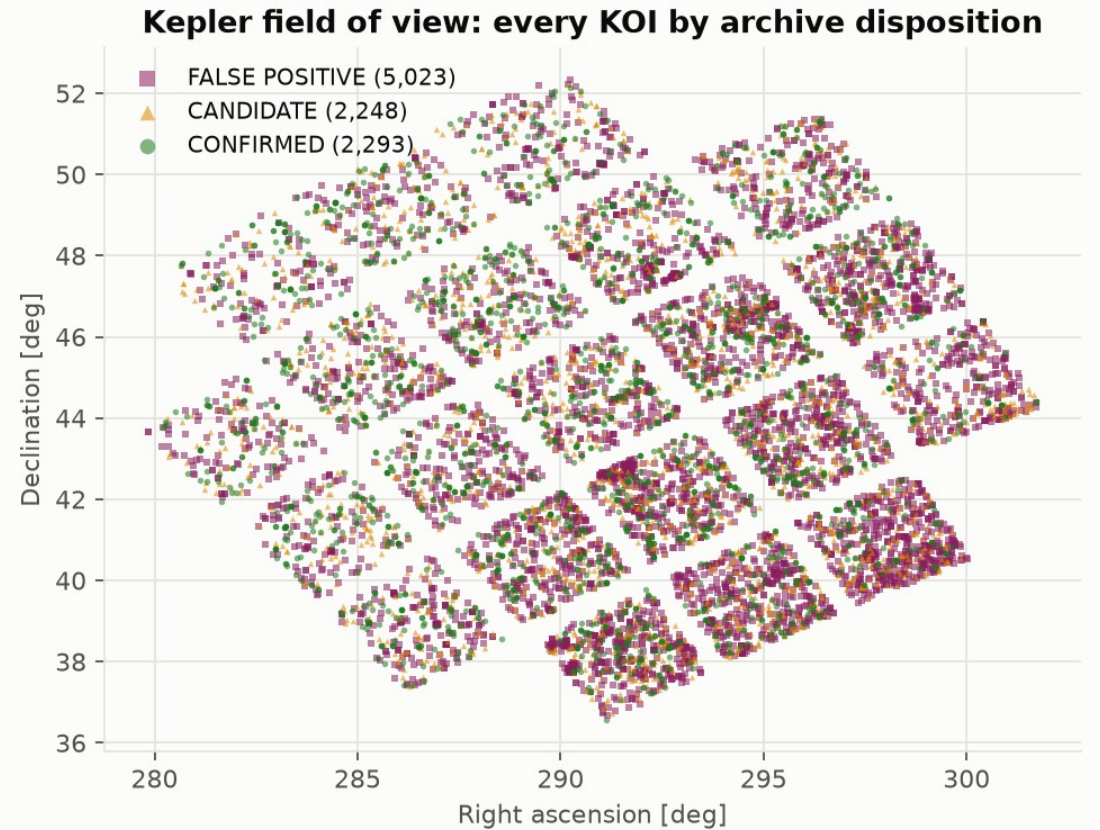
github.com/Rander417/KeplerExoplanet

The app, in your browser (GitHub Pages)

rander417.github.io/KeplerExoplanet

Kepler in one slide

- A space telescope (2009–2018) that stared at ~150,000 stars in Cygnus and Lyra and measured their brightness continuously.
- The transit method: a planet crossing its star dims it slightly and regularly. Earth crossing the Sun would dim it by about 84 parts per million.
- A Kepler Object of Interest (KOI) is a repeating dip promising enough to track. The archive's cumulative table holds 9,564 of them, ids like K00752.01.
- Each KOI carries the archive's verdict: CONFIRMED (a real planet), CANDIDATE (plausible, unsettled) or FALSE POSITIVE (eclipsing stars, a neighbour's light, noise).
- This project asks how far the physics of the signal alone can predict that verdict, and which planets sit in their star's habitable zone.



The 9,564 KOIs on the sky, coloured by verdict (notebook 01)

Three big questions — then and now



Is the KOI an exoplanet?

- 2020: 'GBT and Random Forest reach 90% f1' — with the vetting flags as features
- 2026: the flags are the vetting; physics alone reaches macro f1 0.75. Honest, and still useful



Is it in the habitable zone?

- 2020: 12 confirmed planets, chosen by orbital period and stellar properties
- 2026: 16 small confirmed planets, chosen by the starlight they receive (Kopparapu et al. 2014). The two lists share none



Can future observers use our models?

- 2020: a Flask app on Heroku (offline since 2022)
- 2026: trained on 2020 labels, the model predicted later confirmations with AUC 0.937 — and the app runs in any browser

The data — a 2017-era snapshot and today's table

9,564

KOIs in both tables — the same objects

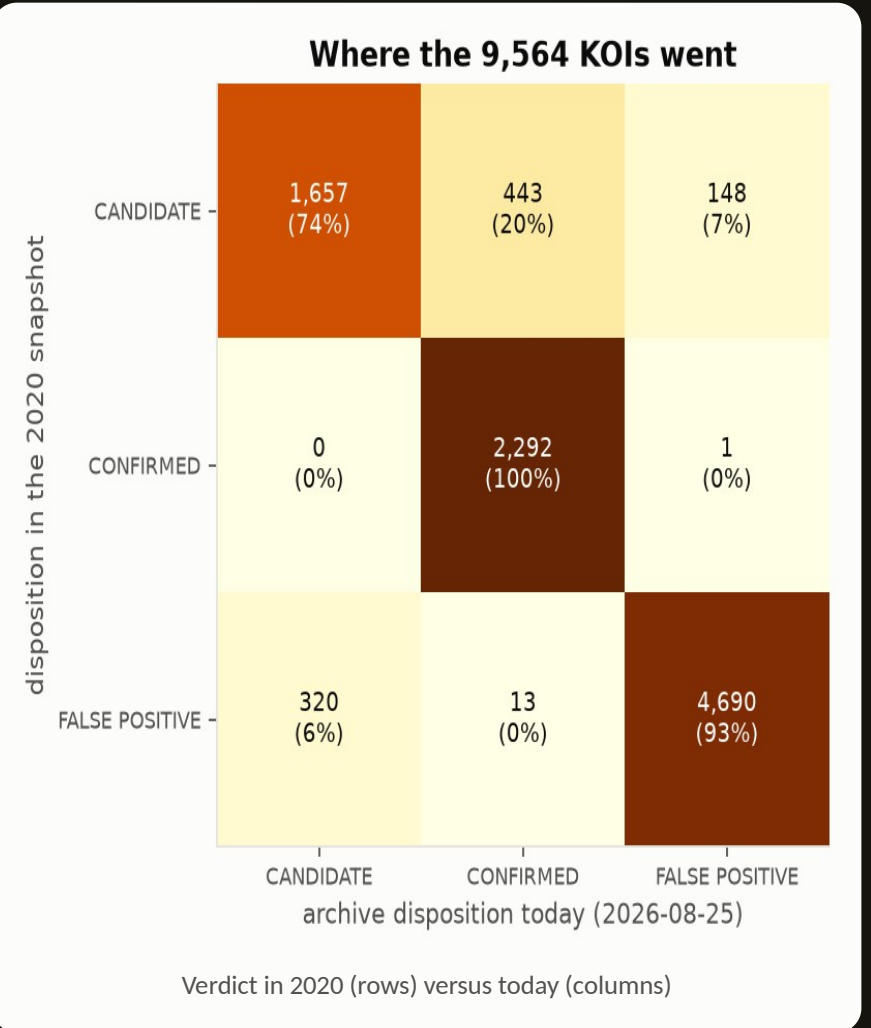
925

verdicts changed since the snapshot (9.7%)

0

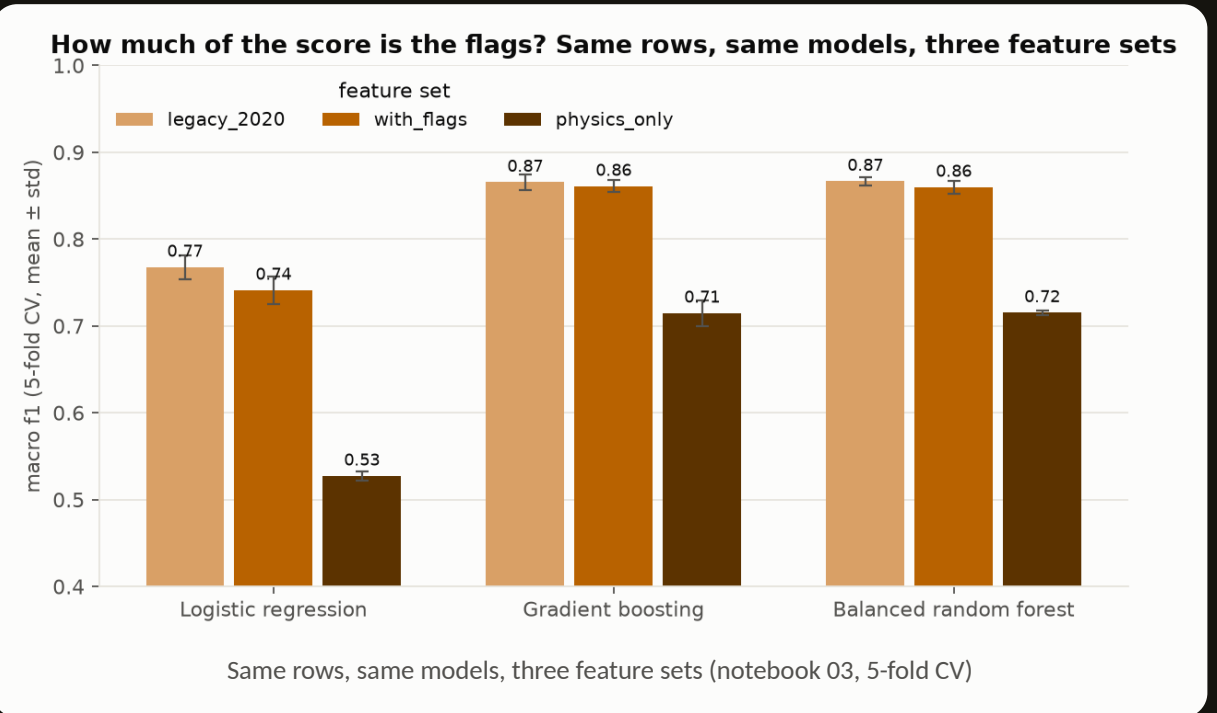
physics columns that changed (depth: rounding only)

- 2020 snapshot: the Kaggle copy of the archive table, 50 columns.
- 2026 pull: the archive's TAP service, all 153 columns, provenance JSON with the query, time and hash.
- 443 candidates were confirmed, 148 dismissed; 320 'false positives' became candidates again; 13 former false positives are confirmed planets today.
- One retraction: Kepler-503 b, CONFIRMED to FALSE POSITIVE.



A tale of two Ys, revisited — the flags are the answer key

- The 2020 models used the four `koi_fpflag_*` vetting flags as features. Any flag set means the pipeline already called it a false positive.
- Between 2020 and today the archive re-decided 925 KOIs — and rewrote the flags with the verdicts (320 'false positives' became candidates with their flags cleared).
- So a flag-fed model learns the vetting logic, not the planets: macro f1 0.86–0.87 with flags versus 0.72 on physics alone, same rows, same models.
- On today's labels the gap is wider still: 0.90 with flags, 0.75 without.
- The refresh reports both variants, always, and builds the app on physics only.



Rule of the refresh: never quote a single f1 without saying which variant — with flags (reproduces the vetting) or physics only (has to earn it).

Cleaning and EDA — the gaps are information

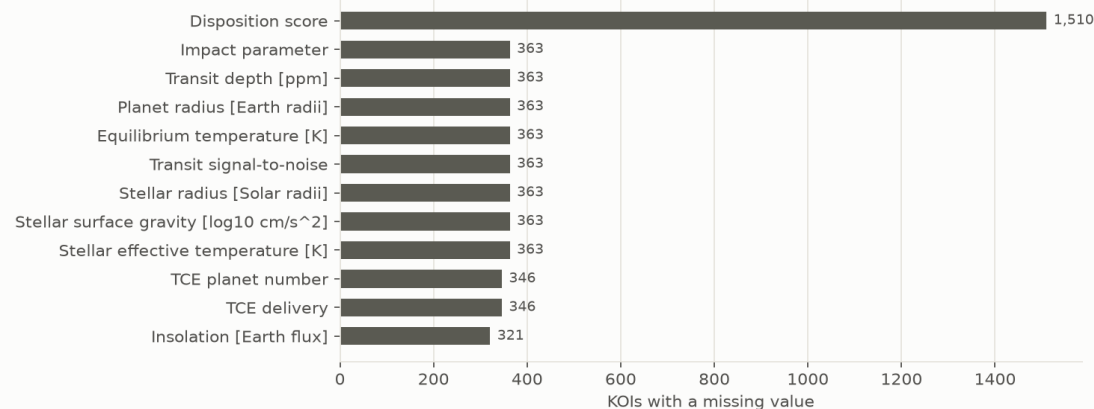
2020

- Dropped the $\pm$ error columns, then the 363 rows with any remaining null.
- Scaled everything and trained on a 75/25 split; scores quoted from one split.
- Imputation tried (mean, median, mode) and rejected — reasonable.

2026

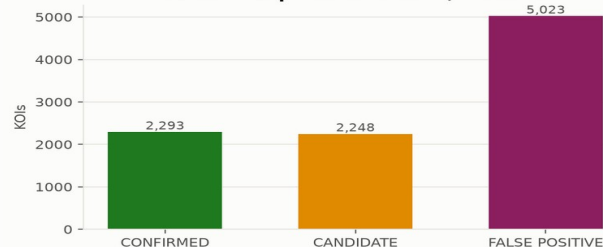
- All 9,564 rows kept: the gaps are not random, they track the verdict (most sit in false positives).
- 'Is this value missing' indicators + NaN-native boosting.
- Every number from 5-fold cross-validation with error bars; nested CV for tuning.

Where the nulls are (kepler_name omitted: 7,271 unnamed KOIs are expected)



Missing values by column (notebook 01)

Archive disposition of the 9,564 KOIs



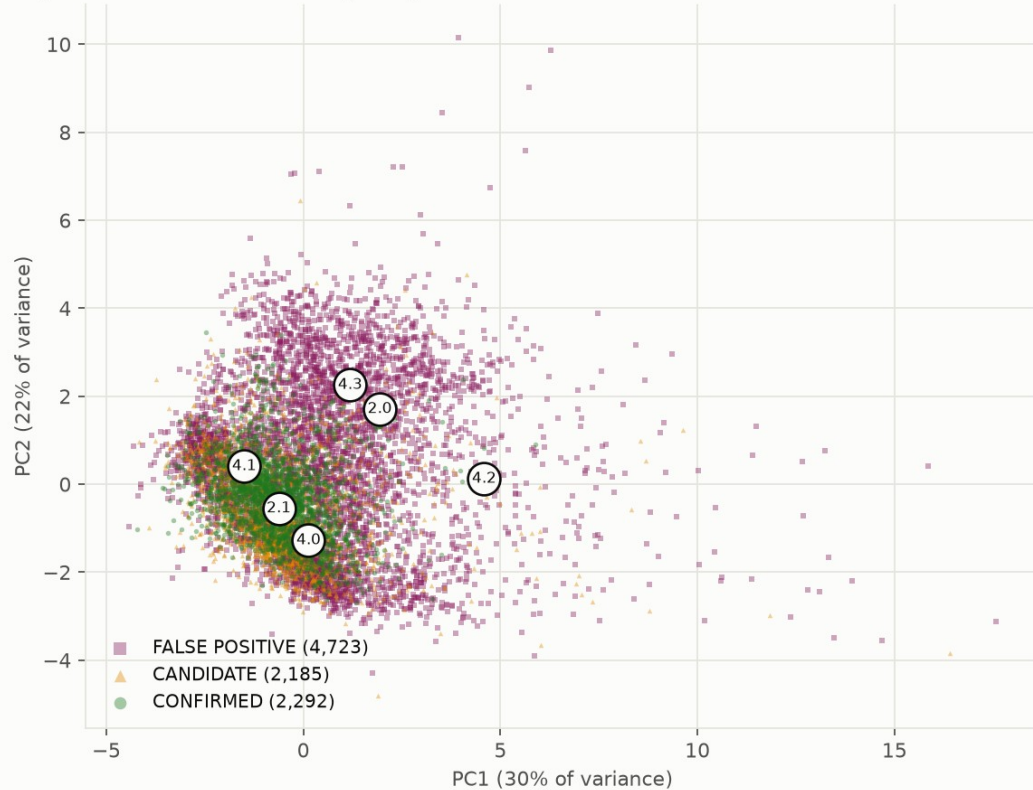
Class balance: the three verdicts

+0.007

macro f1 from keeping the incomplete rows — none, but the catalogue is whole

Clustering — a null result, recorded as one

Physics columns in PCA space; numbered circles are k-means centroids (k.cluster)



Scaled k-means on the physics, projected on two principal components (notebook 02)

2020

- k-means on confirmed planets only, unscaled
- Elbow at 4 (3-D) and 6 (PCA); 'most of the data is consistent'

2026

- All classes, log1p on the heavy tails, standardised
- Agreement with the verdicts: ARI ≤ 0.05 , NMI ≈ 0 for every k

What it means. Unsupervised structure in these twelve measurements does not line up with what the archive decided. That is a finding, not a failure — it is why the supervised models earn their keep.

The models, re-labelled

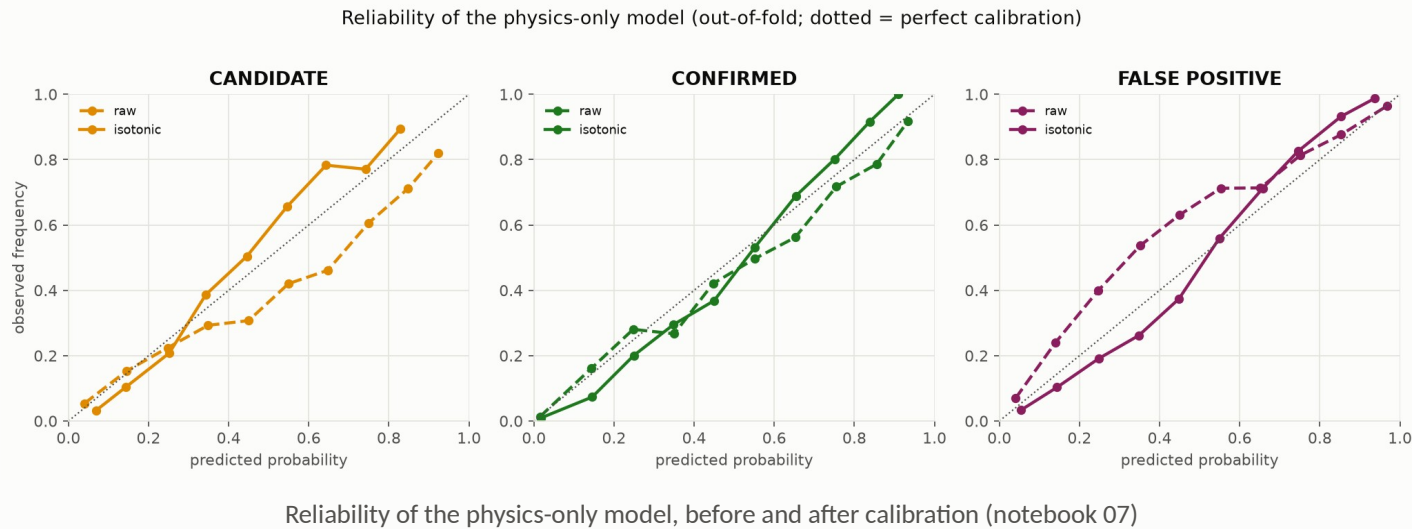
Model	2020 deck said	macro f1, with flags	macro f1, physics only
Logistic regression	83% f1	0.768 ± 0.014	0.527 ± 0.006
Gradient boosting (2020 settings)	90% f1	0.866 ± 0.009	0.715 ± 0.015
Balanced random forest	90% f1	0.867 ± 0.005	0.715 ± 0.003
Deep neural net (2020, Keras)	84% f1	0.79 (one split)	not run yet — Phase 4
Tuned boosting, all rows, nested CV	—	0.862 ± 0.006	0.722 ± 0.006 (snapshot labels) 0.747 ± 0.012 (today's labels)

The 2020 deck's '83 / 90 / 90 / 84% f1' were accuracy and weighted f1 on one 75/25 split; macro f1 is the balanced number. Error bars: 5-fold cross-validation.

Why 'macro'? It averages the three verdicts' f1 equally, so the rare class counts as much as the common one. Accuracy and weighted f1 flatter a model that is good at the big class.

The physics-only ceiling sits at 0.72–0.75 in every construction we tried: different models, all rows, tuning on inner folds, calibration. Confirmation is a follow-up process; the physics cannot see it.

Probabilities you can take literally, and a rule you can read



The label rule

$P(\text{planet-like}) = P(\text{CANDIDATE}) + P(\text{CONFIRMED})$

below 0.5 $\rightarrow$ FALSE POSITIVE

otherwise $\rightarrow$ the larger of CONFIRMED / CANDIDATE

The threshold is a slider in the app; no hidden argmax.

0.84 / 0.55 / 0.84

f1 for CONFIRMED / CANDIDATE / FALSE POSITIVE (out-of-fold, today's labels)

0.921

AUC, planet-like vs false positive

0.534 $\rightarrow$ 0.512

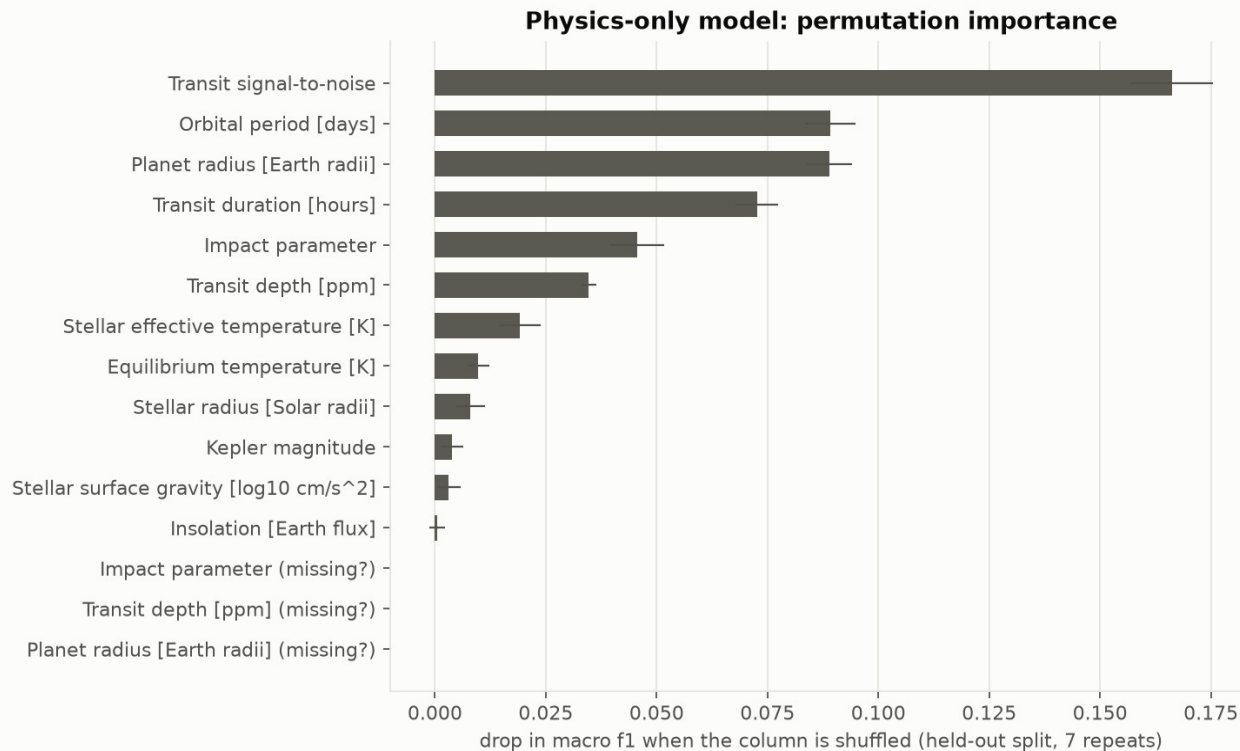
log loss, before $\rightarrow$ after calibration

0.85 / 0.82

precision / recall for 'planet-like' at 0.5

Physics separates planets from false positives well; it separates confirmed from candidate only moderately — confirmation is a follow-up-and-statistics process, not a measurement.

What the physics-only model actually looks at

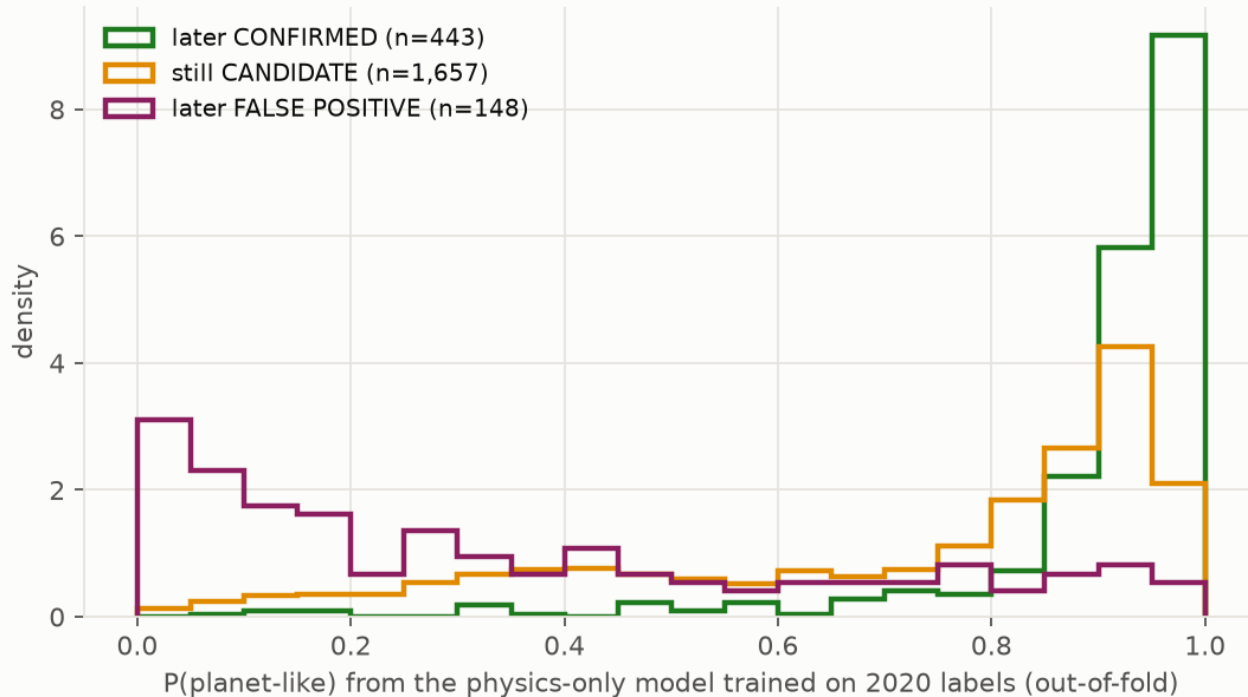


Permutation importance, physics-only model (notebook 07)

- Transit signal-to-noise leads by a wide margin: a clean, deep, repeated dip is the strongest single clue.
- Orbital period and planet radius come next: very large 'planets' and very short periods are mostly eclipsing stars.
- Transit duration and impact parameter carry shape information: grazing crossings and V-shaped dips point to stars, not planets.
- Stellar temperature, gravity and radius matter less on their own — the planet radius already folds the star's size in.
- Twelve columns, all measurable from the light curve and the star: nothing that encodes a human's verdict.

The out-of-time test — question III, answered honestly

2020 candidates: what the physics said then, what the archive says now



What the 2020-trained model thought of the 2020 candidates, split by what happened next (notebook 08)

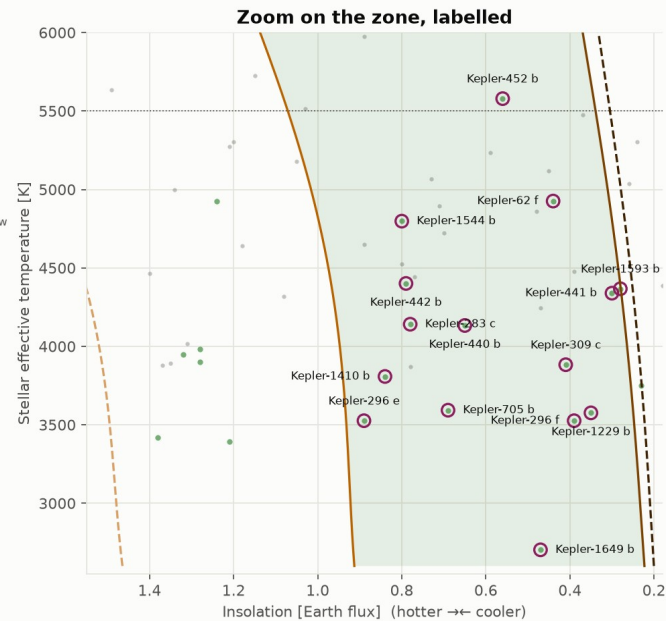
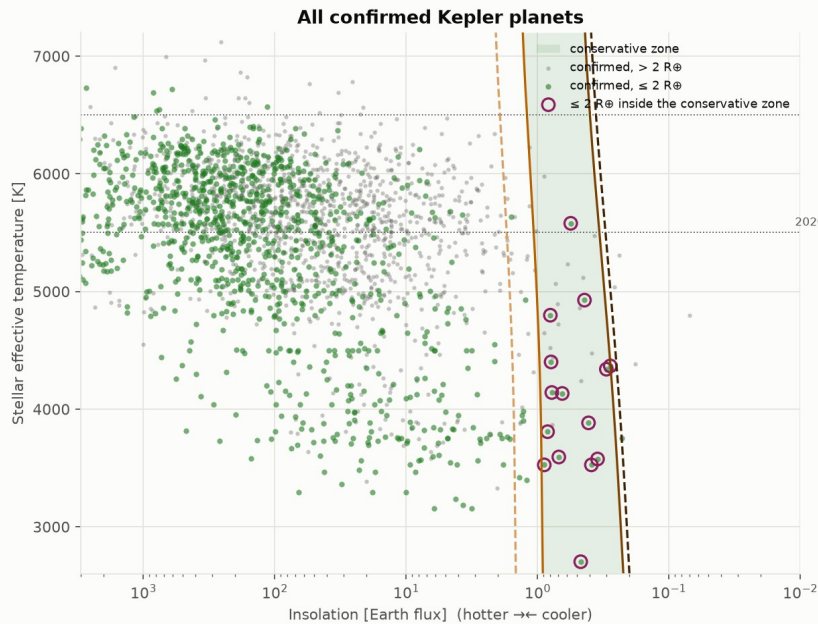
AUC 0.937

later-confirmed vs later-dismissed, on verdicts the model never saw

- 2,248 KOIs were CANDIDATES in 2020. Since then the archive confirmed 443 and dismissed 148.
- The model, trained on 2020 labels, had rated the later-confirmed ones far more planet-like (median 0.95) than the later-dismissed ones (median 0.26).
- That is a forecast checked against the future — the closest thing to 'can future observers use this?'
- The vetting flags could not have passed this test: they were rewritten along with the verdicts.

The habitable zone, done on insolation

Confirmed Kepler planets on the habitable-zone diagram (Kopparapu et al. 2014 limits)



Circled: confirmed planets $\leq 2 R_{\oplus}$ in the conservative zone (notebook 05; today's table adds Kepler-1652 b)

16

confirmed planets $\leq 2 R_{\oplus}$ in the conservative zone today (9 of them $\leq 1.6 R_{\oplus}$, plausibly rocky)

0 of 12

of the 2020 list survive: period and star properties are not the planet's climate

Method. Kopparapu et al. 2014 give the insolation at which a rocky planet loses its water (runaway greenhouse) or freezes (maximum greenhouse) as a function of the star's temperature; Rogers 2015 finds most planets above $1.6 R_{\oplus}$ are not rocky. Caveat: the table's stellar parameters are Kepler DR25 — Kepler-1649 c sits outside and its exo-Venus sibling 1649 b inside on those numbers. A screen, not a verdict.

The app — one file, three ways to run it

Kepler KOI explorer

9,564 Kepler Objects of Interest, their NASA Exoplanet Archive verdicts as of 2026-08-25, a physics-only model that never sees the vetting flags, and the habitable-zone screen. Every number here was computed by a script you can run; nothing is trained in the app. [Source and notebooks](#).

Start here if exoplanets are new to you (two minutes)

Catalogue KOI explorer Habitable zone Model & honesty About

Browse the KOIs

How to read this tab. Filter the 9,564 KOIs, then see them on a period-versus-size map: each dot is one KOI, coloured by today's archive verdict; hover for its name and the model's probability. The table lists the same rows and the download takes exactly what you filtered.

Archive verdict (today) ☐ Habitable zone ☐ Max planet radius (Earth radii) ☐ Only KOIs whose verdict changed since 2020 ☐

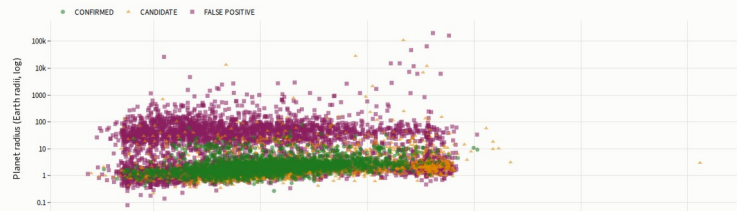
CONFIRMED x CANDIDATE x FALSE POSITIVE x

any no limit Find by KOI or Kepler name

KOI0752.01, Kepler-452...

KOIs shown 9,564 Confirmed 2,748 Candidate 1,977 False Positive 4,839

Period versus radius, coloured by today's archive verdict



Catalogue: filter, plot, download

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9,564 Kepler Objects of Interest, their NASA Exoplanet Archive verdicts as of 2026-08-25, a physics-only model that never sees the vetting flags, and the habitable-zone screen. Every number here was computed by a script you can run; nothing is trained in the app. [Source and notebooks](#).

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One KOI, the model's view, and what-if sliders

How to read this tab. Pick one KOI to see its measurements, what the archive decided, what the model thinks, and whether it sits in its star's habitable zone. The sliders at the bottom let you change the physics and watch the probabilities respond.

Pick a KOI (type to search) KOI016.01 — Kepler-452 b — CONFIRMED

KOI016.01 — Kepler-452 b

- Archive verdict today: CONFIRMED
- Verdict in the 2020 snapshot: CONFIRMED
- Kepler pipeline verdict: CANDIDATE (score 0.77)
- Host star: Teff 5,579 K, radius 0.80 R_☉, Kepler mag 13.4

	value	unit	what it means
Orbital period	384.8476	days	How long one orbit takes: the time between dips
Impact parameter	0.059		How central the crossing is: 0 = straight across the star
Transit duration	9.969	hours	How long each dip lasts, first contact to last
Transit depth	189.9	ppm	How much starlight is blocked, in parts per million
Planet radius	1.09	Earth radii	The planet's size, from the depth and the star's size
Equilibrium temperature	220	K	A no-atmosphere estimate of the planet's temperature
Insolation	0.56	Earth = 1	Starlight received compared with Earth. The habitable zone is between 0.95 and 1.37
Transit signal-to-noise	12.3		How clearly the dip stands out from the noise. Hubble can detect a signal-to-noise of 7
Stellar temperature	5579	K	The star's surface temperature. The Sun is 5,772 K
Stellar surface gravity	4.58	log g	Gravity at the star's surface (log scale). About 4.7 for the Sun

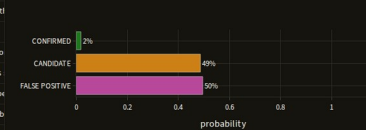
The model's out-of-fold view (the honest number)

Planet-like threshold for the verdict rule 0.50

Planet size ☐ Rule verdict ☐ Archive says ☐

50% CANDIDATE CONFIRMED

Calibrated probabilities (physics only)



Habitable-zone check

Kepler-452 b is in the habitable zone (the Sun's habitable zone is between 0.95 and 1.37 AU)

KOI explorer, dark mode: probabilities and what-if sliders

- Five tabs: Catalogue, KOI explorer, Habitable zone, Model & honesty, About.
- Runs locally, on Streamlit Community Cloud, or entirely in your browser (GitHub Pages).
- The model travels as plain numbers and is evaluated with numpy — the same model everywhere.
- Plain-language help on every control; light and dark themes.

Try Kepler-452 b. The archive says CONFIRMED; the physics-only model says 50/50. Mullally et al. 2018 argued the planet 'can not be confirmed using a purely statistical validation approach'. Sometimes a coin flip is the honest answer.

What we learned

1

Leakage settles arguments

The vetting flags were rewritten with the verdicts. Any score that uses them is a score for copying.

2

The ceiling is real

Physics alone tops out near 0.75 macro f1 in every construction. Confirmation is a process the measurements cannot see.

3

Forecasts beat fits

Trained on 2020, the model predicted the archive's later confirmations with AUC 0.937. That is the number to trust.

4

Ask the planet, not the star

Habitable-zone screens belong on insolation with temperature-dependent limits. The 2020 list of 12 had zero survivors.

5

Reproducibility is the deliverable

Locked environment, tracked data with hashes, 41 tests, every number regenerable — and an app anyone can open.

Tools, credits, references

Tools

Python 3.12 · uv (locked environment)
pandas · numpy · scikit-learn 1.9
plotly · matplotlib · Streamlit
stlite (Streamlit in the browser) · GitHub
Pages
NASA Exoplanet Archive TAP service
pytest (41 tests) · ruff · JupyterLab
Obsidian vault inside the repository

How it was built

A tag team: Claude (Cowork) authored notebooks, code and notes; Claude Code tested, committed and pushed; Rich set direction and reviewed every batch.

References

Kopparapu, R. K. et al. 2014, ApJL 787, L29 — habitable-zone insolation limits vs stellar temperature (arXiv:1404.5292)
Rogers, L. A. 2015, ApJ 801, 41 — 'the majority of 1.6 Earth-radius planets are too low density to be comprised of Fe and silicates alone' (arXiv:1407.4457)
Mullally, F. et al. 2018, AJ 155, 210 — Kepler-452 b 'should be considered a candidate planet' (arXiv:1803.11307)
Vanderburg, A. et al. 2020, ApJL 893, L27 — Kepler-1649 c, rescued from false-positive status (arXiv:2004.06725)
Angelo, I. et al. 2017, AJ 153, 162 — Kepler-1649 b, an exo-Venus (arXiv:1704.03136)
NASA Exoplanet Archive — KOI cumulative table, column definitions and TAP service

Credits

2020 team (Columbia Engineering data-analytics bootcamp): Rich Anderson — ML pipeline; Damien Corr, Priscilla Lin, Tom Greff. Original repo: tom-jj-G/KeplerExoplanets. 2026 refresh: Rich Anderson with Claude.
An educational analysis of public data, not a product of NASA or the Exoplanet Archive. Verdicts are the archive's; probabilities are ours.